## Acidity of Phenol

Phenol is more acidic than aliphatic alcohols. Unlike alcohols it reacts with bases like sodium hydroxide to form sodium phenoxide. This explains the acidic behaviour of phenol.let us consider the aqueous solution of phenol in which the following equilibrium exists.

$$C_6H_5 - OH + H.OH \rightleftharpoons C_6H_5 - O^{\ominus} + H_3O^{\oplus}$$

$K_a$ value for the above equilibrium is $1\times10^{-10}$ at $25^{o}C$. This Ka value indicates that it is more acidic than aliphatic alcohols. This increased acidic behaviour can be explained on the basis of the stability of phenoxide ion. We have already learnt in XI standard that the phenoxide is more stabilised by resonance than phenol.

In substituted phenols, the electron withdrawing groups such as $-NO_2$,-Cl enhances the acidic nature of phenol especially when they are present at ortho and para positions. In such cases,there is a possibility for the extended delocalisation of negative charge on the phenoxide ion. On the other hand the alkyl substitued phenols show a decreased acidity due to the electron releasing +I effect of alkyl group.

**Table: $pK_a$ Values of some alcohols and phenols**

| S.No. | Compound | $pK_a$ Value |
|---|---|---|
| 1 | methanol | 15.5 |
| 2 | ethanol | 15.9 |
| 3 | propan – 2- ol | 16.5 |
| 4 | 2 – methyl propan 2 - ol | 18.0 |
| 5 | Cyclohexanol | 18.0 |
| 6 | Phenol | 10.0 |
| 7 | o – nitrophenol | 7.2 |
| 8 | p – nitrophenol | 7.1 |
| 9 | m - nitrophenol | 8.3 |
| 10 | o – cresol | 10.2 |
| 11 | m – cresol | 10.1 |
| 12 | p – cresol | 10.2 |



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